

OBSERVATORIYA SANGLOK, Tajikistan

WMO: 388440

Lat: 38.2614N

Lon: 69.2175E

Elev: 7346

StdP: 11.19

Time Zone: 5.00 (E05)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.1	11.2	-8.5	4.6	16.1	-5.9	5.3	18.0	17.5	24.5	15.8	24.8	6.8	50	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	78.2	57.6	76.4	56.4	74.4	55.2	59.4	74.5	57.9	73.4	56.7	72.6	4.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.1	79.1	70.3	50.8	72.6	68.2	49.0	67.7	66.3	29.8	73.7	28.5	74.2	27.6	72.7	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.7	12.8	11.3	DB	4.2	84.6	5.2	4.7	0.4	88.0	-2.6	90.8	-5.5	93.4	-9.2	96.9	(4)
(5)			WB	3.2	63.2	5.2	3.4	-0.5	65.7	-3.5	67.7	-6.4	69.6	-10.2	72.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.3	24.7	25.2	35.8	45.0	54.0	63.8	69.3	66.4	59.5	47.9	34.2	28.9		
	DBStd	16.74	5.13	8.29	6.39	6.10	5.05	5.47	3.62	3.40	4.79	6.82	7.73	6.38		
	HDD50	3360	784	695	442	167	19	1	0	0	3	120	475	654		
	HDD65	7071	1249	1115	907	601	341	83	8	23	171	531	923	1119		
	CDD50	2027	0	0	1	16	143	416	599	508	287	55	2	0		
	CDD65	260	0	0	0	0	0	47	142	66	5	0	0	0		
	CDH74	495	0	0	0	0	1	60	338	94	2	0	0	0		
	CDH80	23	0	0	0	0	0	5	15	3	0	0	0	0		
(14) Wind	WSAvg	5.2	6.2	6.0	5.8	5.6	5.1	4.6	4.3	4.1	4.1	4.6	5.6	6.0		
(15) Precipitation	PrecAvg	20.4	2.1	2.8	3.3	3.1	2.4	1.0	0.6	0.3	0.3	0.9	1.7	2.0		
	PrecMax	29.0	3.8	4.8	5.3	7.4	4.7	2.1	1.5	0.8	0.7	2.0	4.4	3.9		
	PrecMin	12.8	0.9	0.9	0.9	1.0	0.4	0.4	0.2	0.1	0.0	0.2	0.5	0.3		
	PrecStd	4.0	0.8	0.9	1.1	1.2	1.1	0.4	0.3	0.2	0.2	0.4	1.0	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.8	47.6	54.7	63.0	69.2	79.7	82.2	78.6	72.8	66.6	56.5	49.7		
		MCWB	28.6	34.6	40.6	47.2	53.0	61.0	59.9	57.2	54.2	46.2	43.3	33.9		
	2%	DB	36.9	42.4	50.4	58.7	66.4	75.1	79.5	76.5	70.2	63.1	50.7	43.9		
		MCWB	27.8	31.8	38.5	45.4	52.1	55.6	58.6	56.5	52.4	45.4	38.1	31.9		
	5%	DB	34.1	38.6	47.5	56.3	64.2	73.2	77.4	74.5	68.3	60.0	47.0	40.6		
		MCWB	25.9	29.8	36.7	43.8	50.7	54.8	57.1	55.2	51.3	43.8	35.6	30.2		
	10%	DB	31.1	35.2	44.7	53.6	62.3	71.4	75.9	72.7	66.7	57.4	43.5	37.5		
		MCWB	24.4	28.2	35.3	42.6	49.1	54.2	56.1	54.6	50.2	42.8	32.9	28.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.8	36.6	43.0	50.1	55.6	61.2	62.4	59.9	56.9	49.1	44.2	36.4		
		MCDB	34.9	44.6	48.2	57.4	66.6	76.5	79.4	75.7	70.6	61.8	54.6	45.1		
	2%	WB	28.8	33.5	40.3	47.5	53.3	58.9	60.4	57.8	54.5	46.6	40.8	33.6		
		MCDB	33.9	41.7	46.6	54.4	63.7	71.5	77.3	73.8	68.2	60.0	49.0	42.4		
	5%	WB	27.4	30.8	38.3	45.8	51.7	56.9	58.4	56.8	52.7	44.9	37.5	31.1		
		MCDB	32.1	36.0	45.4	52.0	62.0	70.5	74.4	72.8	66.4	58.0	43.7	37.7		
	10%	WB	25.8	29.1	36.7	44.6	50.2	55.1	57.2	55.5	51.3	43.5	34.9	29.3		
		MCDB	30.2	34.0	43.0	50.8	60.3	69.1	74.3	71.2	65.0	55.8	40.4	35.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	10.5	9.3	10.0	11.0	11.1	11.2	11.2	11.7	11.0	8.8	8.4		
		MCDBR	12.3	13.6	12.1	13.8	12.9	11.8	13.0	12.8	12.7	13.7	12.6	11.7		
		MCWBR	8.6	9.5	9.2	9.7	9.3	8.1	9.0	8.4	7.9	7.7	8.4	7.4		
	5% WB	MCDBR	10.6	11.4	10.7	11.7	11.9	12.3	12.7	11.9	12.2	13.2	12.0	10.4		
		MCWBR	8.4	9.2	9.3	10.2	9.5	9.6	9.7	8.8	8.8	8.7	9.9	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.288	0.322	0.364	0.370	0.378	0.399	0.395	0.370	0.342	0.358	0.330	0.294		
	taud		2.363	2.258	2.170	2.206	2.214	2.136	2.140	2.217	2.264	2.149	2.213	2.313		
	Ebn at Noon		283	285	282	287	286	279	279	284	286	266	260	270		
	Edh at Noon		28	36	43	44	45	49	48	43	39	39	32	28		
(44) All-Sky Solar Radiation	RadAvg		669	871	1244	1668	2065	2425	2459	2267	1887	1283	778	596		
	RadStd		77	132	110	177	155	109	77	75	61	116	103	74		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	N/A	N/A	N/A	N/A	-0.65	-1.22	N/A	N/A	N/A	N/A

Nomenclature: See separate page